Substrate Framework: Falsifiable Observational Predictions

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A live audit of the substrate-framework's quantitative predictions across particle, gravitational, and cosmological observables. Each predicted value is paired with the current best measurement and the deviation, anchored on three independent reading paths (CODATA m_e, CODATA m_μ [3], and the framework's own Higgs vev ≈ 246.21965 GeV). All values computed at mp.prec = 500. Status flags: † tension, ‡ open prediction, †† regime-tension. As of 2026-05-05, no entry is in falsifying tension ($\geq 3\sigma$) with observation. Three entries (ρ_Λ, H_0 tension magnitude, TOV bound) are flagged ‡ pending closure of upstream prediction work; 38 distinct observables are auto-flagged blue (estimate-pending); 14 half-integer-slot dark-mode candidates are auto-flagged violet (mass predicted, but the framework cannot yet decide whether the slot is occupied). Cell counts in Tables 1–2 exceed observable counts because each observable is rendered across three anchor columns. This document is intended as a falsification target.

I. METHODOLOGY

Predictions are computed from a single mass-anchor parameter h_{eff} that is independently fixed three ways: from CODATA m_e [3], from CODATA m_μ, and from the framework's own Higgs vev. The three anchor reading paths agree to $\sim$ 158 ppt (max pairwise relative deviation across the full 42-particle catalogue), with the spread carried mostly by the e-anchor-vs-vev-anchor pair ($\sim$ 158 ppt) and the μ-anchor-vs-vev-anchor pair ($\sim$ 153 ppt); the e-anchor-vs-μ-anchor pair agrees at $\sim$ 4 ppt. All computations are performed at mp.prec = 500 in mpmath; no floating-point compounding occurs.

a. What each anchor column means. The three anchor columns in Tables 1 and 2 are not three independent measurements; each carries a specific epistemic status:

- The **electron-anchor** column inherits the electron-mass CODATA central value as its base reading; every prediction in that column is propagated from that single experimental input. The column's residual against any other observable is therefore bounded below by the CODATA m_e experimental uncertainty (and above by the framework's structural precision).
- The **muon-anchor** column does the same with the muon-mass CODATA central value as its single experimental input.
- The **vev-anchor** column is qualitatively different: it carries the framework's geometry-intrinsic prediction of the Higgs vev as its base reading, with no CODATA particle-mass input. The values in that column are pure framework predictions; their residual against observation is not bounded below by any experimental uncertainty band on a measured anchor.

A reader comparing the three columns is therefore comparing two experimentally-anchored readings against one geometry-only reading; agreement of all three to $\sim$ 4 ppt is consistency between two independent CODATA inputs and one parameter-free geometric prediction.

b. Auto-flag legend (appears on every cell). Each predicted value's deviation against the observed value is automatically compared against the observation's published experimental uncertainty σ_{obs} (covering $\sim$ 70 observables):

- ✓ **green:** $|\text{dev}| \leq 1\,\sigma_{\text{obs}}$ — prediction sits within the experimental band.
- • **orange:** $1\,\sigma_{\text{obs}} < |\text{dev}| \leq 3\,\sigma_{\text{obs}}$ — marginal.
- × **red:** $|\text{dev}| > 3\,\sigma_{\text{obs}}$ — outside the experimental band; falsifying tension if reproducible.
- ? **blue:** estimate-pending; the prediction depends on upstream work that is not yet closed, so the deviation is *not yet falsifiable*. The work-remaining tally (§ Work remaining) names which item gates each blue prediction.
- ? **violet:** mass predicted, but the framework cannot yet decide whether this candidate slot is physically occupied. Used for spin-$\frac{1}{2}$ half-integer-slot dark-mode candidates.
- plain (no colour): σ_{obs} unavailable for this observable; status undetermined.

II. CROSS-ANCHOR CONSISTENCY

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The framework's mass scale h_{eff} can be fixed from any one of several *dimensionally distinct* observables. The seven anchors used in Table I and the companion data file are not seven variations on the same input; they span a range of physical dimensions:

- m_e — mass, $[M]$ (CODATA $m_e c^2$ in MeV);
- v_C — mass, $[M]$ (G_F-derived Higgs vev in GeV);

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- $m_P c^2$ — mass, $[M]$ at the Planck scale (CODATA, in GeV);
- G — $[L^3 M^{-1} T^{-2}]$ (CODATA Newton's constant in SI units);
- $\lambda_{C,\tau}$ — length, $[L]$ (PDG-derived Compton wavelength of the tau in fm);
- $\alpha_{\rm EM}$ — dimensionless (CODATA fine-structure constant);
- $v_C^{\rm struct}$ — mass, $[M]$ (the framework's own Higgs vev, computed from E_8 invariants alone with *no* PDG input; included as a 7th column in the data file).

A consistency claim across these inputs is therefore not a normalisation property: an internally-rigid framework reproduces a mass-dimension prediction equally well from a length-dimension input, a coupling-strength input, and a gravitational input.

a. Two distinct empirical claims. The cross-anchor exercise establishes *rigidity* (anchor independence) and is logically separate from *accuracy* (the match against observation, addressed by the prediction tables in Sections III–IV). Both must hold for the framework to be correct; here we report the rigidity result, and rely on the prediction tables for the accuracy result.

b. Rigidity result, operational. For every one of the 102 predicted observables in the framework's structural catalogue, the seven anchor-derived predictions agree to *better than 2.53 ppm* (max pairwise spread), with a median pairwise spread of 1.27 ppm. This consistency is a structural property of the framework's equations: no anchor is privileged, no observed value is tuned, and the residual spread reflects the precision of the underlying CODATA/PDG inputs (PDG m_τ at $\sim 6.8 \times 10^{-5}$ and CODATA G at $\sim 2.2 \times 10^{-5}$ dominate) rather than tunable parameters. The parameter-free $v_C^{\rm struct}$ anchor reproduces the mass-anchored predictions at < 1 ppb spread against $v_C^{\rm obs}$, so the rigidity result holds even when the seventh column carries no measured input at all.

c. Why this isn't dimensional analysis through known physics. A reader might ask: "if every anchor produces the same prediction, isn't that just because the anchors are related through known physics?" The framework's structural equations do *not* route through CODATA conversions. Each anchor enters its own geometry-intrinsic relation: m_e via the gateway-slot R-ratio ($m_e = R_e \cdot h_{\rm eff}$ with $R_e = (\dim -2)/\dim$), v_C via the chain-position ratio R_v, m_P via the structural factor $F_{\rm struct} = m_e/m_P$ derived from E_8 invariants alone, G via $F_{\rm struct}^2$ entering as a closed geometric identity, $\lambda_{C,\tau}$ via the tau slot's subharmonic position R_τ, and $\alpha_{\rm EM}$ via the lepton-vacuum-polarisation-corrected $\alpha_2 \sin^2\theta_W$ combination. The fact that all seven paths converge on the same predicted value is a property of the framework's structural identities (companion paper for the full derivation), not of unit conversions; SI dimensional analysis cannot, on its own, relate G to m_e to a part-per-million.

TABLE I: Cross-anchor consistency grid. Each row uses one observable, set to its CODATA/PDG observed value, as the sole input. The remaining five columns show the framework's prediction for every other anchor, derived structurally from that input. Boldface diagonal entries are exact by construction (the anchor's $h_{\rm eff}$ reproduces the anchor itself). Off-diagonal entries show the predicted value above the relative deviation against the observed value beneath. Mass anchors (m_e, v_C, m_P, G, $\lambda_{C,\tau}$) all fix the framework's overall scale $h_{\rm eff}$, so they reproduce each other to within the experimental uncertainty of the anchor input plus the framework's structural identities — the off-diagonal mass-column residuals (typically ~ 1 ppm–2.5 ppm; ~ 150 ppb for the $\lambda_{C,\tau}$ pairs) reflect the published $\sigma_{\rm obs}$ of each anchor (PDG m_τ at $\sim 6.8 \times 10^{-5}$ dominates; G at $\sim 2.2 \times 10^{-5}$ next). The fine-structure constant $\alpha_{\rm EM}$ is a purely *dimensionless* structural prediction (lepton-vacuum-polarisation-corrected $\alpha_2 \sin^2\theta_W$); it carries no scale information, so the $\alpha_{\rm EM}$-row's mass-column entries (marked [*]) reproduce the framework's structural mass scale rather than constituting an independent prediction from $\alpha_{\rm EM}$, and the $\alpha_{\rm EM}$-column entries identical across rows are the framework's single structural prediction $\alpha_{\rm EM} \approx 7.30 \times 10^{-3}$ vs CODATA 7.297×10^{-3} ($\Delta \sim +877$ ppm; lepton-only VP running, hadronic VP not yet substrate-derived — this is a genuine open prediction tracked in J7). Computed at mp.prec = 500.

Anchor used	m_e [MeV]	v_C [GeV]	$m_P c^2$ [GeV]	G [$\mathrm{m^3 kg^{-1} s^{-2}}$]	$\lambda_{C,\tau}$ [fm]	$\alpha_{\rm EM}$ [dim.less]
Observed (CODATA/PDG)	0.51099895069	246.21965	1.22089e+19	6.6743e-11	0.69777139	0.0072973525
m_e anchor	**0.51099895069**	246.21965031 Δ=+1.27 ppb	1.22088858e+19 Δ=−1.160 ppm	6.67431689e-11 Δ=+2.531 ppm	0.697771442 Δ=+74.59 ppb	0.0073037493 Δ=+876.584
v_C anchor	0.51099895004 Δ=−1.27 ppb	**246.21965**	1.22088858e+19 Δ=−1.161 ppm	6.67431691e-11 Δ=+2.533 ppm	0.697771443 Δ=+75.86 ppb	0.0073037493 Δ=+876.584
$m_P c^2$ anchor	0.51099954353 Δ=+1.160 ppm	246.21993597 Δ=+1.161 ppm	**1.22089e+19**	6.67430141e-11 Δ=+210.64 ppb	0.697770633 Δ=−1.086 ppm	0.0073037493 Δ=+876.584
G anchor	0.51099959735 Δ=+1.265 ppm	246.2199619 Δ=+1.267 ppm	1.22089013e+19 Δ=+105.32 ppb	**6.6743e-11**	0.697770559 Δ=−1.191 ppm	0.0073037493 Δ=+876.584
$\lambda_{C,\tau}$ anchor	0.51099898881 Δ=+74.59 ppb	246.21966868 Δ=+75.86 ppb	1.22088867e+19 Δ=−1.086 ppm	6.6743159e-11 Δ=+2.382 ppm	**0.69777139**	0.0073037493 Δ=+876.584
$\alpha_{\rm EM}$ anchor	0.51099895008 [*] Δ=−1.19 ppb	246.21965002 [*] Δ<1 ppb	1.22088858e+19 [*] Δ=−1.161 ppm	6.67431691e-11 [*] Δ=+2.533 ppm	0.697771443 [*] Δ=+75.78 ppb	0.0073037493 Δ=+876.584

[*] The fine-structure constant $\alpha_{\rm EM}$ is dimensionless and $h_{\rm eff}$-independent; it cannot fix the framework's mass scale. Mass-column entries in the $\alpha_{\rm EM}$-anchor row reproduce the framework's structural Higgs-vev mass scale and are not independent predictions from $\alpha_{\rm EM}$.

Full machine-readable cross-anchor grid. A complete observable-by-anchor cross-tabulation covering all 102 predicted observables (every entry in the framework's structural catalogue — 42 particle-mass entries plus electroweak couplings, mixing angles, CP phases, gauge bosons, lifetimes, decay widths, Compton wavelengths, atomic constants, Yukawa couplings, dark-matter candidate spectrum, and gravitational sector) is provided as `cross_anchor_full.csv` (permanent reference, pinned to git tag `v2026-05-05-cross-anchor`). The CSV exposes the seven anchors listed above (including the parameter-free $v_C^{\rm struct}$ column not shown in Table I), and for each observable carries a `max_pairwise_anchor_spread_ppm` column so the rigidity property is visible directly in the data without recomputation. All values are computed at mp.prec = 500. The latest (untagged) revision tracks the `main` branch.

III. SM MASS SPECTRUM, ELECTROWEAK SECTOR, MIXING

TABLE II: Substrate-framework predictions for all derivable Standard Model and gravitational observables, computed at mp.prec = 500 from three independent anchor scales: CODATA electron mass, CODATA muon mass, and the directly-computed structural Higgs vev $v_C = (\dim -2)/(1-\delta_W)\cdot(1+5\alpha_\tau^3/8)$. Each cell carries the predicted value (15-digit precision) with the deviation against the observed value beneath, in the same cell. Auto-flag legend: ✓ green ($|\text{dev}| \le 1\sigma_{\rm obs}$), • orange ($1\sigma < |\text{dev}| \le 3\sigma$), × red ($|\text{dev}| > 3\sigma$), ? blue (estimate-pending; the prediction depends on upstream prediction work that is not yet closed, so the deviation is not yet falsifiable), ? violet (mass predicted, but the framework cannot yet decide whether this candidate slot is physically occupied; used for half-integer-slot dark-mode candidates). Observed column carries the published experimental uncertainty (± ppm/ppb/ppt) below each value where available. Anchor variance (max pairwise ∼ 158 ppt between anchors across the full 42-particle catalogue, carried by the e-anchor-vs-vev-anchor pair; e-anchor-vs-μ-anchor agrees at ∼ 4 ppt) is exposed rather than hidden.

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
CHARGED LEPTON MASSES				
electron (MeV)	0.5109989506900 ±300.000 ppt	0.5109989506900 ✓ ∼ 0 (FP floor)	0.5109989500022 × − 1.346 ppb	0.5109989500806 × − 1.193 ppb
muon (MeV)	105.6583755000 ±22.000 ppb	105.6583756422 ✓ + 1.346 ppb	105.6583755000 ✓ ∼ 0 (FP floor)	105.6583755162 ✓ + 153.386 ppt
tau (MeV)	1776.860000000 ±68.000 ppm	1776.859741241 ✓ − 145.627 ppb	1776.859738849 ✓ − 146.973 ppb	1776.859739121 ✓ − 146.820 ppb
QUARK MASSES				
up (MeV)	2.160000000000 ±170000.000 ppm	2.172135021340 ✓ + 5618.065 ppm	2.172135018416 ✓ + 5618.064 ppm	2.172135018749 ✓ + 5618.064 ppm
down (MeV)	4.670000000000 ±70000.000 ppm	4.666472674001 ✓ − 755.316 ppm	4.666472667719 ✓ − 755.317 ppm	4.666472668435 ✓ − 755.317 ppm
strange (MeV)	93.40000000000 ±92000.000 ppm	93.25272887842 ✓ − 1576.779 ppm	93.25272875289 ✓ − 1576.780 ppm	93.25272876720 ✓ − 1576.780 ppm
charm (MeV)	1273.000000000 ±3600.000 ppm	1274.297662316 ✓ + 1019.373 ppm	1274.297660601 ✓ + 1019.372 ppm	1274.297660796 ✓ + 1019.372 ppm
bottom (MeV)	4183.000000000 ±1700.000 ppm	4178.924491965 ✓ − 974.303 ppm	4178.924486340 ✓ − 974.304 ppm	4178.924486981 ✓ − 974.304 ppm
top (GeV)	172.5700000000 ±1700.000 ppm	172.5698835074 ✓ − 675.046 ppb	172.5698832751 ✓ − 676.392 ppb	172.5698833015 ✓ − 676.238 ppb
NEUTRINO MASSES				
neutrino 1 (meV)	4.510000000000e-10 ±50000.000 ppm	4.516090897055e-10 ✓ + 1350.531 ppm	4.516090890976e-10 ✓ + 1350.530 ppm	4.516090891669e-10 ✓ + 1350.530 ppm
neutrino 2 (meV)	8.613900000000e-9 ±50000.000 ppm	8.625379540353e-9 ✓ + 1332.676 ppm	8.625379528743e-9 ✓ + 1332.675 ppm	8.625379530066e-9 ✓ + 1332.675 ppm
neutrino 3 (meV)	5.014980000000e-8 ±50000.000 ppm	5.021285714219e-8 ✓ + 1257.376 ppm	5.021285707461e-8 ✓ + 1257.374 ppm	5.021285708231e-8 ✓ + 1257.375 ppm
ELECTROWEAK SECTOR				
Higgs vev (GeV)	246.2196500000 ±24.000 ppb	246.2196503134 ✓ + 1.273 ppb	246.2196499819 ✓ − 73.361 ppt	246.2196500197 ✓ + 80.025 ppt
Higgs boson (GeV)	125.2000000000 ±880.000 ppm	125.1997815035 ✓ − 1.745 ppm	125.1997813350 ✓ − 1.747 ppm	125.1997813542 ✓ − 1.746 ppm
W boson (GeV)	80.36920000000 ±160.000 ppm	80.36920047933 ✓ + 5.964 ppb	80.36920037115 ✓ + 4.618 ppb	80.36920038347 ✓ + 4.771 ppb
Z boson (GeV)	91.18760000000 ±23.000 ppm	91.19003493128 ? + 26.702 ppm	91.19003480853 ? + 26.701 ppm	91.19003482252 ? + 26.701 ppm

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Table II continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
Weinberg mixing fraction	0.2232000000000 ±140.000 ppm	0.2232441517887 ? + 197.813 ppm	0.2232441517887 ? + 197.813 ppm	0.2232441517887 ? + 197.813 ppm
SU(2) coupling	0.6533000000000 ±1500.000 ppm	0.6528252345176 ? − 726.719 ppm	0.6528252345176 ? − 726.719 ppm	0.6528252345176 ? − 726.719 ppm
$U(1)_Y$ coupling (EW norm)	0.3500000000000 ±1000.000 ppm	0.3499811975896 ? − 53.721 ppm	0.3499811975896 ? − 53.721 ppm	0.3499811975896 ? − 53.721 ppm
$U(1)_Y$ coupling (GUT norm)	0.4519000000000 ±1000.000 ppm	0.4518237832501 ? − 168.658 ppm	0.4518237832501 ? − 168.658 ppm	0.4518237832501 ? − 168.658 ppm
SU(2) fine-structure	0.03394000000000 ±1500.000 ppm	0.03391438943683 ? − 754.583 ppm	0.03391438943683 ? − 754.583 ppm	0.03391438943683 ? − 754.583 ppm
$U(1)_Y$ fine-structure (EW)	0.009745000000000 ±2000.000 ppm	0.009747192918717 ? + 225.030 ppm	0.009747192918717 ? + 225.030 ppm	0.009747192918717 ? + 225.030 ppm
$U(1)_Y$ fine-structure (GUT)	0.01624100000000 ±2000.000 ppm	0.01624532153119 ? + 266.088 ppm	0.01624532153119 ? + 266.088 ppm	0.01624532153119 ? + 266.088 ppm
EM fine-structure (tree)	0.007575408000000 ±1.000 ppb	0.007571189103257 ? − 556.920 ppm	0.007571189103257 ? − 556.920 ppm	0.007571189103257 ? − 556.920 ppm
LEPTON YUKAWA COUPLINGS				
electron Yukawa	2.935028322980e-6 ±300.000 ppt	2.935028319245e-6 × − 1.273 ppb	2.935028319245e-6 × − 1.273 ppb	2.935028319245e-6 × − 1.273 ppb
muon Yukawa	0.0006068707660433 ±22.000 ppb	0.0006068707660878 ✓ + 73.361 ppt	0.0006068707660878 ✓ + 73.361 ppt	0.0006068707660878 ✓ + 73.361 ppt
tau Yukawa	0.01020576347354 ±68.000 ppm	0.01020576197431 ✓ − 146.900 ppb	0.01020576197431 ✓ − 146.900 ppb	0.01020576197431 ✓ − 146.900 ppb
QUARK YUKAWA COUPLINGS				
up Yukawa	1.240640742819e-5 ±170000.000 ppm	1.247610742106e-5 ✓ + 5618.064 ppm	1.247610742106e-5 ✓ + 5618.064 ppm	1.247610742106e-5 ✓ + 5618.064 ppm
down Yukawa	2.682311235631e-5 ±70000.000 ppm	2.680285239466e-5 ✓ − 755.317 ppm	2.680285239466e-5 ✓ − 755.317 ppm	2.680285239466e-5 ✓ − 755.317 ppm
strange Yukawa	0.0005364622471263 ±92000.000 ppm	0.0005356163642517 ✓ − 1576.780 ppm	0.0005356163642517 ✓ − 1576.780 ppm	0.0005356163642517 ✓ − 1576.780 ppm
charm Yukawa	0.007311739192631 ±3600.000 ppm	0.007319192575631 ✓ + 1019.372 ppm	0.007319192575631 ✓ + 1019.372 ppm	0.007319192575631 ✓ + 1019.372 ppm
bottom Yukawa	0.02402592697783 ±1700.000 ppm	0.02400251842267 ✓ − 974.304 ppm	0.02400251842267 ✓ − 974.304 ppm	0.02400251842267 ✓ − 974.304 ppm
top Yukawa	0.9911915416122 ±1700.000 ppm	0.9911908712511 ✓ − 676.318 ppb	0.9911908712511 ✓ − 676.318 ppb	0.9911908712511 ✓ − 676.318 ppb
HIGGS SELF-COUPLING				
Higgs self-coupling	0.1292805654113 ±1800.000 ppm	0.1292801138469 ✓ − 3.493 ppm	0.1292801138469 ✓ − 3.493 ppm	0.1292801138469 ✓ − 3.493 ppm
STRONG COUPLING				
strong coupling	0.1180000000000 ±7600.000 ppm	0.1180478309232 ? + 405.347 ppm	0.1180478309232 ? + 405.347 ppm	0.1180478309232 ? + 405.347 ppm
PMNS NEUTRINO MIXING				
PMNS solar mixing fraction	0.3030000000000 ±50000.000 ppm	0.3065788699420 ? + 11811.452 ppm	0.3065788699420 ? + 11811.452 ppm	0.3065788699420 ? + 11811.452 ppm
PMNS atmospheric mixing fraction	0.5720000000000 ±30000.000 ppm	0.5714285714286 ? − 999.001 ppm	0.5714285714286 ? − 999.001 ppm	0.5714285714286 ? − 999.001 ppm
PMNS reactor mixing fraction	0.02203000000000 ±30000.000 ppm	0.02197802197802 ? − 2359.420 ppm	0.02197802197802 ? − 2359.420 ppm	0.02197802197802 ? − 2359.420 ppm
PMNS CP phase (rad)	3.438298626429 ±50000.000 ppm	3.455751918949 ? + 5076.142 ppm	3.455751918949 ? + 5076.142 ppm	3.455751918949 ? + 5076.142 ppm
CKM QUARK MIXING				
Cabibbo mixing fraction	0.05064000000000 ±4000.000 ppm	0.05035971223022 ? − 5534.909 ppm	0.05035971223022 ? − 5534.909 ppm	0.05035971223022 ? − 5534.909 ppm
CKM atmospheric mixing fraction	0.001682000000000 ±15000.000 ppm	0.001681796794019 ? − 120.812 ppm	0.001681796794019 ? − 120.812 ppm	0.001681796794019 ? − 120.812 ppm
CKM reactor mixing fraction	1.460000000000e-5 ±30000.000 ppm	1.452605387890e-5 ? − 5064.803 ppm	1.452605387890e-5 ? − 5064.803 ppm	1.452605387890e-5 ? − 5064.803 ppm
CKM CP phase (rad)	1.143190660056 ±50000.000 ppm	1.151917306316 ? + 7633.588 ppm	1.151917306316 ? + 7633.588 ppm	1.151917306316 ? + 7633.588 ppm
QCD THETA ANGLE				
QCD theta angle (rad)	0.0	0.0	0.0	0.0
MASSLESS GAUGE BOSONS				
photon (MeV)	0.0	0.0	0.0	0.0
gluon 1 (MeV)	0.0	0.0	0.0	0.0
gluon 2 (MeV)	0.0	0.0	0.0	0.0
gluon 3 (MeV)	0.0	0.0	0.0	0.0
gluon 4 (MeV)	0.0	0.0	0.0	0.0
gluon 5 (MeV)	0.0	0.0	0.0	0.0

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Table II continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
gluon 6 (MeV)	0.0	0.0	0.0	0.0
gluon 7 (MeV)	0.0	0.0	0.0	0.0
gluon 8 (MeV)	0.0	0.0	0.0	0.0
GRAVITATIONAL SECTOR				
Newton's G ($m^3kg^{-1}s^{-2}$)	6.674300000000e-11 ±22.000 ppm	6.674316892333e-11 ✓ + 2.531 ppm	6.674316910301e-11 ✓ + 2.534 ppm	6.674316908253e-11 ✓ + 2.533 ppm
Planck mass (GeV)	1.220890000000e+22 ±11.000 ppm	1.220888583580e+22 ✓ − 1.160 ppm	1.220888581936e+22 ✓ − 1.161 ppm	1.220888582124e+22 ✓ − 1.161 ppm
ANOMALOUS MAGNETIC MOMENTS				
electron anomalous moment	0.001159652180590 ±1.700 ppt	0.001160667181068 ? + 875.263 ppm	0.001160667181068 ? + 875.263 ppm	0.001160667181068 ? + 875.263 ppm
muon anomalous moment	0.001165920590000 ±460.000 ppb	0.001166944225438 ? + 877.963 ppm	0.001166944225438 ? + 877.963 ppm	0.001166944225438 ? + 877.963 ppm
tau anomalous moment	0.001177210000000 ±2100.000 ppm	0.001176416612358 ? − 673.956 ppm	0.001176416612358 ? − 673.956 ppm	0.001176416612358 ? − 673.956 ppm
FERMI CONSTANT				
Fermi constant (GeV^{-2})	1.166378700000e-5 ±500.000 ppb	1.166378704555e-5 ✓ + 3.905 ppb	1.166378707695e-5 ✓ + 6.597 ppb	1.166378707337e-5 ✓ + 6.291 ppb
LEPTON LIFETIMES				
muon lifetime (s)	2.196981100000e-6 ±1.000 ppm	2.197000755680e-6 ? + 8.947 ppm	2.197000758638e-6 ? + 8.948 ppm	2.197000758301e-6 ? + 8.948 ppm
tau lifetime (s)	2.903000000000e-13 ±400.000 ppm	2.975066649707e-13 ? + 24824.888 ppm	2.975066653712e-13 ? + 24824.889 ppm	2.975066653255e-13 ? + 24824.889 ppm
W AND Z DECAY WIDTHS				
W total decay width (GeV)	2.085000000000 ±2000.000 ppm	2.097035189882 ? + 5772.273 ppm	2.097035187059 ? + 5772.272 ppm	2.097035187381 ? + 5772.272 ppm
Z total decay width (GeV)	2.495500000000 ±90.000 ppm	2.508962911473 ? + 5394.875 ppm	2.508962908096 ? + 5394.874 ppm	2.508962908481 ? + 5394.874 ppm
Z invisible width (3 neutrinos) (GeV)	0.4995000000000 ±3000.000 ppm	0.4976876525197 ? − 3628.323 ppm	0.4976876518498 ? − 3628.325 ppm	0.4976876519261 ? − 3628.324 ppm
Z leptonic width (per lepton) (GeV)	0.08398000000000 ±500.000 ppm	0.08389802836406 ? − 976.085 ppm	0.08389802825113 ? − 976.087 ppm	0.08389802826400 ? − 976.086 ppm
HIGGS DECAY WIDTHS				
Higgs width to bb (GeV)	0.002200000000000 ±140000.000 ppm	0.002303033293435 ? + 46833.315 ppm	0.002303033296517 ? + 46833.317 ppm	0.002303033296166 ? + 46833.316 ppm
Higgs width to tau tau (GeV)	0.0002300000000000 ±100000.000 ppm	0.0002591191587633 ? + 126605.038 ppm	0.0002591191584145 ? + 126605.037 ppm	0.0002591191584543 ? + 126605.037 ppm
Higgs width to cc (GeV)	0.0001200000000000 ±300000.000 ppm	0.0001140529646162 ? − 49558.628 ppm	0.0001140529647697 ? − 49558.627 ppm	0.0001140529647522 ? − 49558.627 ppm
Higgs total fermionic width (GeV)	0.002510000000000 ±50000.000 ppm	0.002676205416815 ? + 66217.298 ppm	0.002676205419701 ? + 66217.299 ppm	0.002676205419372 ? + 66217.299 ppm
COMPTON WAVELENGTHS AND ATOMIC CONSTANTS				
electron Compton wavelength (fm)	2426.310235380 ±1.500 ppb	2426.310234659 ✓ − 296.967 ppt	2426.310237925 ✓ + 1.049 ppb	2426.310237553 ✓ + 895.687 ppt
muon Compton wavelength (fm)	11.73444106000 ±22.000 ppb	11.73444108357 ✓ + 2.009 ppb	11.73444109937 ✓ + 3.355 ppb	11.73444109757 ✓ + 3.201 ppb
tau Compton wavelength (fm)	0.6977713900000 ±70.000 ppm	0.6977714420463 ✓ + 74.589 ppb	0.6977714429855 ✓ + 75.935 ppb	0.6977714428785 ✓ + 75.782 ppb
Bohr radius (fm)	52917.72105000 ±1.500 ppb	52871.37485602 ? − 875.816 ppm	52871.37492719 ? − 875.815 ppm	52871.37491908 ? − 875.815 ppm
Rydberg energy (eV)	13.60569312299 ±14.000 ppt	13.62955663377 ? + 1753.936 ppm	13.62955661542 ? + 1753.934 ppm	13.62955661751 ? + 1753.934 ppm
EFFECTIVE NEUTRINO SPECIES				
effective neutrino species	3.046000000000 ±56000.000 ppm	3.000000000000 ? − 15101.773 ppm	3.000000000000 ? − 15101.773 ppm	3.000000000000 ? − 15101.773 ppm
DARK MATTER ULTRALIGHT BAND (predicted-but-unobserved; falsification targets)				
DM ultralight 1 (heaviest) (eV)	–	6.205011605723e-7 ? predicted	6.205011597370e-7 ? predicted	6.205011598322e-7 ? predicted
DM ultralight 2 (eV)	–	2.232018563210e-9 ? predicted	2.232018560205e-9 ? predicted	2.232018560548e-9 ? predicted
DM ultralight 3 (eV)	–	8.028843752553e-12 ? predicted	8.028843741745e-12 ? predicted	8.028843742977e-12 ? predicted
DM ultralight 4 (eV)	–	2.888073292285e-14 ? predicted	2.888073288398e-14 ? predicted	2.888073288841e-14 ? predicted

(continued on next page)

Table II continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
DM ultralight 5 (eV)	–	1.038875284995e-16 ? predicted	1.038875283596e-16 ? predicted	1.038875283756e-16 ? predicted
DM ultralight 6 (eV)	–	3.736961456815e-19 ? predicted	3.736961451785e-19 ? predicted	3.736961452358e-19 ? predicted
DM ultralight 7 (lightest) (eV)	–	1.344230739862e-21 ? predicted	1.344230738052e-21 ? predicted	1.344230738258e-21 ? predicted
DARK MATTER MeV–GeV BAND (predicted-but-unobserved; direct-detection targets)				
DM MeV–GeV mode A (MeV)	–	0.8335357327932 ? predicted	0.8335357316712 ? predicted	0.8335357317991 ? predicted
DM MeV–GeV mode B (MeV)	–	5.828295554901 ? predicted	5.828295547056 ? predicted	5.828295547950 ? predicted
DM MeV–GeV mode C (MeV)	–	46.62636443921 ? predicted	46.62636437645 ? predicted	46.62636438360 ? predicted
DM MeV–GeV mode D (GeV)	–	1.492043662055 ? predicted	1.492043660046 ? predicted	1.492043660275 ? predicted
DM MeV–GeV mode E (GeV)	–	4.816169319819 ? predicted	4.816169313336 ? predicted	4.816169314075 ? predicted
DM MeV–GeV mode F (GeV)	–	592.3496679011 ? predicted	592.3496671038 ? predicted	592.3496671946 ? predicted
HALF-SLOT CANDIDATES (spin-1/2 dark-mode candidates; slot occupancy open)				
DM half-slot candidate 1 (MeV)	–	0.6552852647454 ? slot open	0.6552852638634 ? slot open	0.6552852639639 ? slot open
DM half-slot candidate 2 (MeV)	–	2.775832926115 ? slot open	2.775832922379 ? slot open	2.775832922805 ? slot open
DM half-slot candidate 3 (MeV)	–	7.267224947661 ? slot open	7.267224937879 ? slot open	7.267224938994 ? slot open
DM half-slot candidate 4 (MeV)	–	11.75861696921 ? slot open	11.75861695338 ? slot open	11.75861695518 ? slot open
DM half-slot candidate 5 (MeV)	–	30.78445888608 ? slot open	30.78445884464 ? slot open	30.78445884936 ? slot open
DM half-slot candidate 6 (MeV)	–	49.81030080294 ? slot open	49.81030073590 ? slot open	49.81030074354 ? slot open
DM half-slot candidate 7 (MeV)	–	130.4050604920 ? slot open	130.4050603164 ? slot open	130.4050603364 ? slot open
DM half-slot candidate 8 (MeV)	–	552.4047008539 ? slot open	552.4047001104 ? slot open	552.4047001951 ? slot open
DM half-slot candidate 9 (MeV)	–	0.6126241048944 ? slot open	0.6126241040698 ? slot open	0.6126241041638 ? slot open
DM half-slot candidate 10 (MeV)	–	1.732762635557 ? slot open	1.732762633224 ? slot open	1.732762633490 ? slot open
DM half-slot candidate 11 (MeV)	–	3.465525271114 ? slot open	3.465525266449 ? slot open	3.465525266980 ? slot open
DM half-slot candidate 12 (MeV)	–	9.801985678311 ? slot open	9.801985665117 ? slot open	9.801985666620 ? slot open
DM half-slot candidate 13 (MeV)	–	13.86210108445 ? slot open	13.86210106580 ? slot open	13.86210106792 ? slot open
DM half-slot candidate 14 (MeV)	–	27.72420216891 ? slot open	27.72420213159 ? slot open	27.72420213584 ? slot open
RULED OUT BY FRAMEWORK				
4th chiral generation	**ruled out:** Framework predicts exactly three chiral fermion generations. **Falsified** if a 4th chiral lepton or quark generation is observed.			

(continued on next page)

Table II continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
Fundamental scalar beyond Higgs	**ruled out:** Framework predicts the Higgs is the unique fundamental scalar. **Falsified** if a non-Higgs fundamental scalar is observed.			
Quantum-gravity scale below $\sim 10\,E_{\rm Planck}$	**ruled out:** Framework predicts no quantum-gravity scale below $\sim 10\,E_{\rm Planck}$. **Falsified** if quantum-gravity signatures appear at sub-Planckian energies.			
Ultralight DM modes below $\sim 10^{-21}$ eV	**ruled out:** Framework predicts no ultralight-DM modes below $\sim 10^{-21}$ eV (Lyman-α structure-formation cutoff, Rogers & Peiris 2021). **Falsified** if direct detection finds a discrete-spectrum DM mode in this range.			
PQ axion (strong CP)	**ruled out:** Framework predicts $\theta_{\rm QCD} = 0$ exactly with no PQ axion required. **Falsified** if $\theta_{\rm QCD}$ is measured nonzero.			
Dark photon (kinetic-mixing operator)	**ruled out:** Framework predicts no 4D kinetic-mixing dark-photon operator. **Falsified** if dark-photon kinetic-mixing-style coupling is detected.			

IV. GRAVITATIONAL AND COSMOLOGICAL OBSERVABLES

TABLE III: Extended framework predictions: gravitational sector, cosmological observables, and additional constants. **Honest scope:** the framework's genuine gravity-sector predictions are exactly two — Newton's G ($\sim$ 2500 ppb) and the Planck mass m_P ($\sim$ 1 ppb). All other gravity-sector entries (Schwarzschild radii, perihelion precessions, light deflection, Shapiro delay, gravitational redshift) are GR consequences computed using the framework's predicted G together with *observed* body parameters ($M_\odot$, $M_\oplus$, $R_\odot$, etc.); they are GR-consistency checks of the framework G, not independent predictions. The MOND a_0 and $\Lambda_{\rm crit-tension}$ rows are literature MOND (Milgrom 1983; Famaey & Binney 2005) treated as ansatz; closure of the underlying equations is open. Computed at mp.prec = 500, anchored on three independent reading paths (CODATA m_e, CODATA m_μ, framework Higgs vev). Each predicted value is shown to 15-digit precision with the deviation against the observed value beneath. The Observed column carries the published experimental uncertainty (± ppm/ppb/ppt) below each value where available. † tension; ‡ open prediction; †† regime-tension; ? pending closure (literature ansatz or GR-consistency check).

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
Higgs vev (structural identity) (GeV)	246.2196500000 ±24.368 ppm	246.2196500197 ✓ + 80.025 ppt	246.2196500197 ✓ + 80.025 ppt	246.2196500197 ✓ + 80.025 ppt
Newton's G ($\mathrm{m^3kg^{-1}s^{-2}}$)	6.674300000000e-11 ±22.474 ppm	6.674316900433e-11 ✓ + 2.532 ppm	6.674316918401e-11 ✓ + 2.535 ppm	6.674316916354e-11 ✓ + 2.535 ppm
Planck mass m_P (GeV)	1.220890000000e+19 ±11.467 ppm	1.220888583580e+19 ✓ − 1.160 ppm	1.220888581936e+19 ✓ − 1.161 ppm	1.220888582124e+19 ✓ − 1.161 ppm
GR perihelion precession (Mercury) (arcsec/century)	43.11000000000 ±4871.260 ppm	42.99184850049 ✓ − 2740.698 ppm	42.99184850049 ✓ − 2740.698 ppm	42.99184850049 ✓ − 2740.698 ppm
GR perihelion precession (Venus) (arcsec/century)	8.626000000000 ±3477.858 ppm	8.626741315799 ✓ + 85.940 ppm	8.626741315799 ✓ + 85.940 ppm	8.626741315799 ✓ + 85.940 ppm
GR perihelion precession (Earth) (arcsec/century)	3.840000000000 ±13020.833 ppm	3.839649160079 ✓ − 91.365 ppm	3.839649160079 ✓ − 91.365 ppm	3.839649160079 ✓ − 91.365 ppm
GR perihelion precession (Mars) (arcsec/century)	1.350000000000 ±37037.037 ppm	1.351262708288 ✓ + 935.339 ppm	1.351262708288 ✓ + 935.339 ppm	1.351262708288 ✓ + 935.339 ppm
GR perihelion precession (Jupiter) (arcsec/century)	0.06230000000000 ±16051.364 ppm	0.06232950642600 ✓ + 473.618 ppm	0.06232950642600 ✓ + 473.618 ppm	0.06232950642600 ✓ + 473.618 ppm
Light deflection (solar limb) (arcsec)	1.751700000000 ±456.699 ppm	1.751644031301 ✓ − 31.951 ppm	1.751644031301 ✓ − 31.951 ppm	1.751644031301 ✓ − 31.951 ppm
Shapiro round-trip (Earth-Saturn conj., $b = 1.6\,R_\odot$) (μs)	260.0000000000 ±76923.077 ppm	264.4693464873 ✓ + 17189.794 ppm	264.4693464873 ✓ + 17189.794 ppm	264.4693464873 ✓ + 17189.794 ppm
Solar grav. redshift z (limb)	2.120000000000e-6 ±4716.981 ppm	2.123052477022e-6 ✓ + 1439.848 ppm	2.123052477022e-6 ✓ + 1439.848 ppm	2.123052477022e-6 ✓ + 1439.848 ppm

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Table III continued

Observable	Observed	electron-anchor	muon-anchor	vev-anchor
Schwarzschild radius (Sun) (km)	2.953000000000 ±338.639 ppm	2.954015216528 • + 343.792 ppm	2.954015216528 • + 343.792 ppm	2.954015216528 • + 343.792 ppm
Schwarzschild radius (Earth) (mm)	8.870000000000 ±112.740 ppm	8.870125332416 ✓ + 14.130 ppm	8.870125332416 ✓ + 14.130 ppm	8.870125332416 ✓ + 14.130 ppm
MOND scale $a_0 = cH_0/(2\pi)$ (m/s^2)	1.2×10^{-10} ±0.02 stat ±0.24 syst	1.0422×10^{-10} (CMB) 1.1288×10^{-10} (local) ? Milgrom 1983; ansatz, open	1.0422×10^{-10} (CMB) 1.1288×10^{-10} (local) ? Milgrom 1983; ansatz, open	1.0422×10^{-10} (CMB) 1.1288×10^{-10} (local) ? Milgrom 1983; ansatz, open
$\Lambda_{\mathrm{crit-tension}} = (H_0/c)^2$ (m^{-2})	1.097121526139e-52	$5.3086e-53$ (CMB) $6.2273e-53$ (local) ? Friedmann factor $3\Omega_\Lambda\sim2$ open	$5.3086e-53$ (CMB) $6.2273e-53$ (local) ? Friedmann factor $3\Omega_\Lambda\sim2$ open	$5.3086e-53$ (CMB) $6.2273e-53$ (local) ? Friedmann factor $3\Omega_\Lambda\sim2$ open
Hubble constant H_0 (km/s/Mpc)	∼ 73 (local) ∼ 67 (CMB)	open †† regime tension; pending	open †† regime tension; pending	open †† regime tension; pending

V. PREDICTED BUT UNOBSERVED (FALSIFICATION TARGETS)

Discrete dark-matter spectrum (21 modes total: 13 with specific mass predictions, 8 high-mass with envelope). The framework predicts a discrete spectrum of dark-matter masses, none yet detected. The 13 specific-mass predictions are:

- *Ultralight band (7 modes):* 0.62 μeV, 2.23 neV, 8.03 peV, 28.9 feV, 0.10 aeV, 0.37 zeV, 1.34 zeV. The Lyman-α-forest ultralight-DM bound $m \gtrsim 2\times10^{-20}$ eV at 95% CL [10] excludes the 1.34 zeV mode; the next-lightest mode (0.37 zeV is below; 0.10 aeV ∼ 19× above the bound) is above the bound.
- *MeV–GeV band (6 modes):* 0.83 MeV, 5.83 MeV, 46.6 MeV, 1.49 GeV, 4.82 GeV, 593 GeV.

The remaining 8 high-mass modes lie within the envelope [191 GeV, 1.1 EeV]; specific masses ‡ open.

Slot-occupancy disclaimer. The dark-matter modes enumerated above are *candidate slots* only. The framework predicts the mass that would be carried by each slot *if* it is occupied; it does not yet predict which slots are physically populated, nor with what relic abundance. We are explicitly *not* predicting that detector experiments will find a particle in every listed mass band. Some, all, or none of these candidate slots may host observable dark-matter species; the framework's slot-occupancy prediction function (which slots are populated and at what abundance) is unfinished work and is gated by the open J-items below. The 14 spin-$\frac{1}{2}$ half-integer-slot modes (auto-flagged violet in the predictions table) are the clearest example of this: each has a definite predicted mass, but whether the half-integer slot can support a populated state at all is a spin-cover question the framework cannot currently answer. Treat the dark-matter spectrum as a list of *permissible* masses for any dark-sector state the framework does admit, not as a list of guaranteed detections.

CP rationals (framework predictions). $\delta_{\mathrm{CP}}^{\mathrm{PMNS}} = 11\pi/10 \approx 198°$ (consistent with NuFit-6.0 [2] normal ordering: at the IC24 fit $\delta_{\mathrm{CP}}^{\mathrm{best}} = 212°$ with asymmetric range $-41°/+26°$, the framework's 198° sits at $\sim 0.3\sigma$ on the low-δ side; awaiting DUNE/HyperK). $\delta_{\mathrm{CKM}} = 11\pi/30 \approx 1.152$ rad (within $\sim 1\sigma$ of the CKMfitter/PDG 2024 central value [1, 7]).

Three matter generations ($N_{\mathrm{gen}} = 3$). The framework predicts exactly three chiral fermion generations.

Strong CP: $\theta_{\mathrm{QCD}} = 0$ exactly, with no PQ axion required.

Internal topology-discrimination test: $|t| = 6.92\sigma$. A measurement-level discrimination test against the leading alternative scenario clears 5σ on both percentile and BCa 95%-CI floors.

No quantum-gravity scale below $\sim 10\,E_{\mathrm{Planck}}$; no fourth chiral generation; no fundamental scalar beyond Higgs.

VI. UPCOMING EXPERIMENTS AND WHAT WE PREDICT THEY WILL FIND

The following named experiments have data releases or sensitivity upgrades scheduled in the 2026–2030 window that directly test the framework's predictions. Each entry pairs the experiment with the specific framework prediction it probes and the central-value expectation we anticipate the data to converge on.

Neutrino sector

a. JUNO (Jiangmen Underground Neutrino Observatory). Status: taking data since August 2025. *First-result window:* 2026–2030. The 59-day exposure already published $\sin^2\theta_{12} = 0.3092 \pm 0.0087$ and $\Delta m^2_{21} = (7.50 \pm 0.12) \times 10^{-5}$ eV^2; the full 6-year program will reach $\sim 0.5\%$ precision on $\sin^2\theta_{12}$. *Framework prediction:* $\sin^2\theta_{12} = 0.3066$. The current measurement sits at $\sim 0.3\sigma$ above the framework value; we expect JUNO's high-statistics result to converge on a central value within $\sim 0.5\%$ of 0.3066.

b. Hyper-Kamiokande. Status: under construction; commissioning 2027–2028; data taking from 2028. *First-result window:* 2029–2030. *Framework prediction:* $\delta_{\rm CP}^{\rm PMNS} = 11\pi/10 \approx 198^\circ$ (normal ordering). Hyper-K's projected 6°–20° precision on $\delta_{\rm CP}$ will discriminate 198° from the current global-fit best fits (177° in NuFit-6.0 IC19, 212° in IC24) at high significance. We anticipate Hyper-K will converge near 198°.

c. IceCube-Upgrade. Status: 7 new strings deploying austral summer 2025–2026. *First-result window:* 2027–2028. *Framework prediction:* $\sin^2\theta_{23} = 4/7 \approx 0.5714$. We anticipate IceCube-Upgrade's atmospheric-oscillation result to land near 0.5714.

d. T2K + NOvA joint analysis. Status: first joint result published October 2025; next combined update expected 2027–2028. The current 3σ band on $\delta_{\rm CP}$ does not exclude CP conservation. *Framework prediction:* same $11\pi/10$ as Hyper-K. We anticipate the next T2K+NOvA joint update will pull the central value toward 198° as data accumulates.

CKM sector

e. LHCb (Run 3 → Upgrade II). Status: Run 3 data ongoing; analysis 2026–2028. The current measurement $\gamma = (64.6 \pm 2.8)^\circ$ is already within 0.5σ of the framework's $11\pi/30 \approx 66.0^\circ$. *First-result window:* Run 3 datasets give $\sim 1^\circ$ precision by 2027–2028; HL-LHC+Upgrade II projects 0.3° by ~ 2036. We anticipate LHCb's central value to converge near 66°.

f. Belle II. Status: running. Independent γ extraction expected $\sim$ 2027 providing cross-check on LHCb. *Framework prediction:* same 66°. Independent confirmation expected.

Cosmology and CMB

g. Simons Observatory. Status: Large Aperture Telescope first light February 2025; two Small Aperture Telescopes running since 2023; full array online mid-2026. *First-result window:* 2027–2028. *Framework prediction:* $n_s = 58/60 \approx 0.9667$. SO's projected $\sigma(n_s) \approx 0.002$ (vs Planck's 0.0042) will discriminate 0.9667 from the current Planck-2018 central value 0.9649 ± 0.0042 at greater than 2σ. We anticipate SO's central value to land near 0.9667.

h. LiteBIRD. Status: JAXA satellite; launch JFY 2027 ($\sim$2028); 3-year survey. Joint constraint on n_s and tensor-to-scalar ratio r at $\sigma(r) \sim 10^{-3}$. Confirms or refutes the n_s prediction in conjunction with SO.

i. Vera C. Rubin Observatory (LSST). Status: operations begin early 2026; alert stream live February 2026; DR1 $\sim$ 2028. SN-Ia+lensing program will deliver an independent H_0 at $\sim 1\%$ precision by 2028–2030. *Framework prediction:* the local-vs-CMB H_0 tension ($\sim 9\%$) is a genuine substrate-physical effect, not measurement error; Rubin's local H_0 should land near the SH0ES central value, $\sim$73 km/s/Mpc.

j. Euclid & DESI. Status: both running; Euclid DR1 expected 2026–2027; DESI 5-year results expected 2027 with extension through 2028. *Framework prediction:* BAO/lensing-derived H_0 should remain in the CMB-extrapolated band ($\sim$ 67–68 km/s/Mpc); neither survey will resolve the $\sim 9\%$ tension to within measurement error. The persistence of the gap is the framework prediction.

k. CMB-S4. Status: revised plan published June 2025; full survey starts mid-2030s. Long-term ultimate test of $n_s = 0.9667$ at $\sigma(r) \sim 5 \times 10^{-4}$ tensor sensitivity.

Dark-matter direct detection

l. LZ (LUX-ZEPLIN). Status: operating; latest world-best WIMP exclusion limits published December 2025. Next major release $\sim$2027. *Framework prediction:* discrete-mass dark-matter peaks at 1.49 GeV, 4.82 GeV, and 593 GeV are within LZ's accessible window. We expect LZ to either detect a peak at one of these specific masses (consistent with the framework) or continue to set null limits across the band.

m. XENONnT. Status: running since 2021; updates 2026–2028. Same framework predictions apply; XENONnT provides an independent cross-check.

n. XLZD (DARWIN successor). Status: Design Book published 2025; construction late 2020s; operations early 2030s. 60–80 tonne LXe target reaches $\sigma_{\rm SI} \sim 3 \times 10^{-49}$ cm^2, covering the framework's MeV–GeV and high-mass-envelope predictions down to the neutrino fog.

o. ADMX (axion direct detection). Status: currently scanning 1.10–1.31 GHz (4.6–5.4 μeV); upgrades extend the range through 2030. *Framework prediction:* no QCD axion (since $\theta_{\rm QCD} = 0$ is a structural identity, not a Peccei-Quinn relaxation), so we predict ADMX will continue to set null limits and *not* detect any DFSZ-class axion signal in any frequency band.

p. NANOGrav, EPTA, SKA (pulsar-timing arrays). Status: NANOGrav 15-year dataset published; SKA-Mid first light 2028; 30-year datasets simulated through the 2030s. *Framework prediction:* the framework's 0.37 zeV and 1.34 zeV ultralight modes lie in the PTA-sensitive 10^{-23}–10^{-22} eV range. We anticipate PTA-program de-

tection of one of these specific zeV-band masses if they are dominant DM components, or continued null limits.

Lepton magnetic moments

q. J-PARC E34 muon $g-2$/EDM. Status: beam commissioning 2025; data taking $\sim$2030; target precision 450 ppb. Independent ultracold-muon technique with different systematics. Cross-checks the closed Fermilab Muon $g-2$ result; framework predicts no discrepancy beyond Standard-Model lattice-QCD evaluations.

r. Belle II tau program. Status: running. Few-$\times 10^{-6}$ sensitivity on a_τ projected by 2028.

Higgs and electroweak precision

s. HL-LHC (ATLAS & CMS). Status: Run 3 finishes $\sim$ 2026; HL-LHC Run 4 begins mid-2030s; 3000 fb^{-1} per detector. Reaches $\kappa_b \sim 3\%$, $\kappa_\tau \sim 2\%$, total $\Gamma_H \sim 5\%$ precision on Higgs partial widths. *Framework predictions:* the predicted partial widths and the observation that no fourth chiral fermion generation exists below $\sim$ 2.5 TeV; we expect HL-LHC to confirm the Standard-Model partial-width pattern and to push 4th-generation exclusion to $\sim$2.5 TeV without discovery. Di-Higgs $\to \lambda_{HHH}$ extraction at 3–5σ tests the no-new-fundamental-scalar prediction.

Gravitational tests

t. LIGO/Virgo/KAGRA O5 & LISA. Status: O5 run starting 2027; LISA launch $\sim$2035. Multi-messenger BNS+GRB timing constrains $\Delta c/c$ down to 10^{-17} and tests vacuum dispersion of gravitational waves. *Framework prediction:* no frame-dependent c, no vacuum dispersion; we anticipate continued null measurement of any deviation from the speed-of-light universality.

u. ACES (Atomic Clock Ensemble in Space). Status: launched April 21, 2025 to ISS. Tests gravitational redshift to $\sim 3 \times 10^{-6}$ and time variation of fundamental constants. *Framework prediction:* no time variation of constants beyond the framework's structural integers, no Lorentz-violation signature.

v. LHC searches for sub-Planckian quantum gravity. Status: ongoing through HL-LHC. *Framework prediction:* no detection of quantum-gravity signatures (e.g. micro-black-hole production) below $\sim 10\, E_{\rm Planck}$.

Anchored summary of near-term decisive tests

The most actionable falsification targets within the 2026–2028 window are:

- **Simons Observatory** on $n_s = 0.9667$ (2027–2028 result, $> 2\sigma$ discrimination from Planck central);
- **JUNO** on $\sin^2\theta_{12} = 0.3066$ (already at 0.3σ in 59-day data; full-precision result by $\sim$2030);
- **LHCb Run 3** on $\gamma = 11\pi/30$ (already at 0.5σ; $\sim 1^\circ$ precision by 2027–2028);
- **Hyper-Kamiokande** on $\delta_{\rm CP}^{\rm PMNS} = 11\pi/10$ (first decisive results 2029–2030);
- **LZ/XENONnT** on the MeV–GeV dark-matter spectrum (continuous updates through 2027).

VII. WHAT WOULD FALSIFY THE FRAMEWORK

The list below is mixed: items marked † are committed and falsifiable today; items marked ‡ become falsifiable once upstream prediction work closes (see § Work remaining).

† Fourth chiral fermion generation observed.

† $\theta_{\rm QCD}$ measured nonzero.

‡ Direct-detection dark-matter peak at a mass either (a) outside the union of the 13 specific-mass predictions $\pm$ predicted width AND outside the high-mass envelope [191 GeV, 1.1 EeV], or (b) inside the high-mass envelope but inconsistent with the 8 high-mass values once those are committed.

‡ $\delta_{\rm CP}^{\rm PMNS}$ measured inconsistent with $11\pi/10$ at $\geq 3\sigma$ in normal ordering. (Currently auto-flagged blue; becomes a committed-falsifiable prediction once the upstream PMNS-mixing work closes.)

† Topology-discrimination null result: a reproducible measurement-level test showing the framework's preferred scenario loses to the leading alternative at $> 5\sigma$.

† Astrophysical detection of an azimuthal mode-pattern on a rotating compact body (e.g. pulsar surface oscillations via NICER or successor X-ray timing missions) whose symmetry group, classified as a finite subgroup of $O(3)$, falls outside $\{\mathbb{Z}_n, D_n, T, O, I\}$.

‡ Stable non-collapsed compact object whose mass exceeds the framework-predicted substrate TOV bound $M_{\rm sub-TOV}$ (value ‡ open; bound is geometric at saturation, $r_{\rm phys} = r_s$).

† CMB-S4 measurement of n_s inconsistent with the framework's prediction $n_s = (N-2)/N$ at the $\Delta n_s \sim 10^{-4}$ level, with $N = 60$ as the framework canonical value ($n_s = 58/60 = 0.9667$ at $N = 60$ vs. $n_s = 56/58 = 0.9655$ at $N = 58$ differ by $\sim 1.2 \times 10^{-3}$, so the N-dependence is itself part of the falsifiable claim).

† Frame-dependent c, vacuum dispersion, or any departure from speed-of-light universality at any energy scale.

† Experimental confirmation of a quantum-gravity scale below $\sim 10\,E_{\rm Planck} \approx 10^{20}$ GeV (via gravitational decoherence at observer-energies $\ll E_{\rm Planck}$, or any single-graviton-exchange process detected at sub-Planckian energies).

VIII. WORK REMAINING (PRIORITIZED BY UNBLOCK COUNT)

The two tables together carry 38 distinct blue-flagged observables that are *not yet falsifiable* because the upstream prediction work that would commit them at structural precision is not yet closed. The list below tallies open work items in priority order, counting how many blue observables each item turns into committed predictions.

a. J1. Lepton/quark mixing structure — unblocks 12 observables. Blocks: PMNS $\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, $\delta_{\rm CP}$ (4); CKM $\sin^2\theta_{12}^{\rm CKM}$, $\sin^2\theta_{23}^{\rm CKM}$, $\sin^2\theta_{13}^{\rm CKM}$, $\delta_{\rm CKM}$ (4); Higgs partial widths $\Gamma_{H\to bb,\tau\tau,cc,ff}$ (4).

b. J2. Gauge-coupling precision — unblocks 9 observables. Blocks: $\alpha_{\rm EM}^{\rm tree}$, α_2, $\alpha_1^{\rm EW}$, $\alpha_1^{\rm GUT}$ (4); g_2, $g_1^{\rm EW}$, $g_1^{\rm GUT}$ (3); $\sin^2\theta_W$ (1); Z boson cascading from the W-mass+$\sin^2\theta_W$ identity (1).

c. J3. $\alpha_{\rm EM}$ hadronic running — unblocks 5 observables. Blocks: lepton anomalous magnetic moments $a_e^{\rm LO}$, $a_\mu^{\rm LO}$, $a_\tau^{\rm LO}$ (3); Bohr radius a_0 (1); Rydberg energy Ry (1).

d. J4. Gravity / cosmology block — unblocks 4 observables. A single closure decomposes into five sub-items: (a) the form of the MOND interpolation function (cf. Famaey–Binney [4] and the radial-acceleration relation [5]); (b) the prefactor in the relation $a_0 = cH_0/(2\pi)$ (cf. Milgrom 1983 [6]); (c) the residual $\sim 50\%$ Friedmann coefficient that closes $\Lambda_{\rm crit-tension} = (H_0/c)^2$ to the full ΛCDM relation; (d) the local-vs-CMB H_0 regime tension at $\sim 9\%$ (SH0ES [9] vs. Planck [8]); (e) the dark-mode spatial density at cluster scales needed for combined-MOND-plus-dark-mode acceleration tests. Blocks: ρ_Λ (dark-tower contribution); MOND a_0; $\Lambda_{\rm crit-tension}$; H_0 regime tension.

e. J5. NLO electroweak radiative corrections — unblocks 4 observables. Blocks: Γ_W, Γ_Z, $\Gamma_Z^{\rm invisible}$, $\Gamma_Z^{\rm leptonic}$ (4).

f. J6. NLO+ QED/QCD corrections for lepton lifetimes — unblocks 2 observables. Blocks: τ_μ, τ_τ (2).

g. J7. $\Lambda_{\rm QCD}$ closure — unblocks 1 observable. Blocks: α_s at lower scales (1).

h. J8. Thermal-history closure — unblocks 1 observable. Blocks: $N_{\rm eff}$ at recombination (1).

i. Total unblock count. J1 (12) + J2 (9) + J3 (5) + J4 (4) + J5 (4) + J6 (2) + J7 (1) + J8 (1) = 38 observables, matching the count of distinct blue-flagged observables across both tables.

j. Strategic note. J1 and J2 together account for 21 of 38 blue observables (55%); J3 and J7 are tightly coupled. Realistically, closing J1+J2+J3+J7 together unblocks 27 of 38 (71%); J4 is structurally independent and is the second-priority closure.

k. ‡-flagged items. ρ_Λ and the H_0 regime tension fall under J4 (sub-items c and d). The TOV bound's nuclear-scale closure is a separate item not yet in the J1–J8 tally; it remains ‡-flagged.

IX. PROVENANCE

All numerical content auto-generated at mp.prec = 500. Source code, per-section reproducibility traces, and JSON output artefacts mirrored at github.com/gable-digital/papers.

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